HSC Mathematics Extension 2: Probability Distributions

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`https://github.com/vuhung16au/math-olympiad-ml`

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The theory of probabilities is at bottom nothing but common sense reduced to calculus; it enables us to appreciate with exactness that which accurate minds feel with a sort of instinct.

Pierre-Simon Laplace

Contents

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1 Introduction

1.1 Project Overview

This booklet is a growing collection of probability distribution problems for NSW HSC Mathematics Extension 2 students. It covers continuous probability distributions (including the normal distribution), binomial distributions, and selected advanced topics such as Monte Carlo simulations and the hypergeometric distribution.

1.2 Target Audience

This resource is written for students who want extra practice with probability distributions in a clear, classroom-friendly format. Full worked solutions are provided for Part 1 problems, with hints for Part 2 problems. Each problem contains a takeaways box summarising the key idea.

1.3 How to Use This Booklet

- Read the introductory theory for each topic before attempting problems.
- Attempt each problem before reading the solution.
- Pay attention to whether a question involves a discrete or continuous distribution.
- Use the takeaways to build pattern recognition for future HSC questions.

1.4 Extension 1 Knowledge Needed

To work through this booklet, students should be comfortable with:

- **Basic probability:** sample spaces, complementary events, conditional probability.
- **Algebra:** solving equations, working with exponentials and logarithms.
- **Calculus:** integration, including integration by substitution.
- **Binomial theorem:** expansion of $(1+x)^n$ and binomial coefficients $\binom{n}{r}$.
- **Statistics basics:** mean, variance, standard deviation.

1.5 Core Formulas and Ideas

Probability density function (pdf). A function $f(x)$ is a probability density function if

$$f(x) \geq 0 \quad \text{for all } x, \qquad \text{and} \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1.$$

The probability that X lies in the interval $[a, b]$ is

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx.$$

Mean (expected value). For a continuous random variable with pdf $f(x)$,

$$\mu = E(X) = \int_{-\infty}^{\infty} x\,f(x)\,dx.$$

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Variance.

$$\sigma^2 = \mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2.$$

Standard deviation. $\sigma = \sqrt{\mathrm{Var}(X)}$.

Standard normal distribution. The standard normal distribution $Z \sim N(0,1)$ has pdf

$$\phi(z) = \frac{1}{\sqrt{2\pi}}\, e^{-z^2/2}.$$

General normal distribution. If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1).$$

Binomial distribution. If $X \sim B(n,p)$, then

$$P(X=r) = \binom{n}{r} p^r (1-p)^{n-r}, \quad r = 0, 1, \ldots, n.$$

The mean is $\mu = np$ and the variance is $\sigma^2 = np(1-p)$.

Normal approximation to binomial. For large n,

$$X \sim B(n,p) \approx N(np,\, np(1-p))\,.$$

Sample proportion. For a sample of size n from a population with proportion p,

$$\hat{p} = \frac{X}{n}, \qquad E(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

2 Continuous Probability Distributions

2.1 Relative Frequency

Relative frequency is the proportion of times an event occurs in a repeated experiment. If an event occurs k times in n trials, its relative frequency is

$$\text{relative frequency} = \frac{k}{n}.$$

As the number of trials n increases, the relative frequency of an event tends towards its theoretical probability. This is the basis of the frequency interpretation of probability.

Remark 2.1

Histograms of relative frequencies approximate continuous probability density functions as the class width decreases and the sample size increases.

2.2 Continuous Distributions

A **continuous random variable** X can take any value in an interval (or union of intervals) of real numbers. Instead of assigning probability to individual values, we use a **probability density function** (pdf) $f(x)$.

Properties of a pdf:

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- $f(x) \geq 0$ for all x.
- $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$.
- $P(a \leq X \leq b) = \displaystyle\int_a^b f(x)\,dx$.

Note that for a continuous random variable, $P(X = c) = 0$ for any particular value c. This means $P(a \leq X \leq b) = P(a < X < b)$.

Cumulative distribution function (CDF). The CDF is defined by

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt.$$

The pdf is the derivative of the CDF: $f(x) = F'(x)$.

2.3 Mean and Variance of a Distribution

Mean (Expected Value). The mean of a continuous random variable X with pdf $f(x)$ is

$$\mu = E(X) = \int_{-\infty}^{\infty} x\,f(x)\,dx.$$

Variance. The variance measures the spread of the distribution:

$$\sigma^2 = \mathrm{Var}(X) = E\big[(X-\mu)^2\big] = \int_{-\infty}^{\infty} (x-\mu)^2\,f(x)\,dx.$$

An equivalent and often more convenient formula is

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2.$$

Standard deviation. $\sigma = \sqrt{\mathrm{Var}(X)}$.

Properties of expectation:

- $E(aX + b) = aE(X) + b$ for constants a, b.
- $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$.

2.4 The Standard Normal Distribution

The **standard normal distribution** is denoted $Z \sim N(0, 1)$. It has mean $\mu = 0$ and standard deviation $\sigma = 1$.

Its pdf is

$$\phi(z) = \frac{1}{\sqrt{2\pi}}\,e^{-z^2/2}, \qquad -\infty < z < \infty.$$

The cumulative distribution function is

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^{z} \phi(t)\,dt.$$

Values of $\Phi(z)$ are read from the standard normal table.

Symmetry property: $P(Z \leq -z) = 1 - P(Z \leq z)$, or equivalently $\Phi(-z) = 1 - \Phi(z)$.

Key values:

- $P(-1 \leq Z \leq 1) \approx 0.6827$
- $P(-2 \leq Z \leq 2) \approx 0.9545$
- $P(-3 \leq Z \leq 3) \approx 0.9973$

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2.5 General Normal Distributions

If $X \sim N(\mu, \sigma^2)$, the pdf of X is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Standardising. To compute probabilities, convert X to the standard normal variable Z:

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1).$$

Then

$$P(a \le X \le b) = P\left(\frac{a-\mu}{\sigma} \le Z \le \frac{b-\mu}{\sigma}\right).$$

The 68-95-99.7 rule. For any normal distribution:

- approximately 68% of values lie within one standard deviation of the mean.
- approximately 95% of values lie within two standard deviations of the mean.
- approximately 99.7% of values lie within three standard deviations of the mean.

2.6 Applications of the Normal Distribution

The normal distribution models many naturally occurring quantities, including:

- heights and weights in a population,
- measurement errors in scientific experiments,
- scores on standardised tests,
- lifetimes of manufactured components.

Finding probabilities. To find $P(X \le k)$ for $X \sim N(\mu, \sigma^2)$:

1. Standardise: compute $z = \dfrac{k-\mu}{\sigma}$.
2. Read $\Phi(z)$ from the standard normal table.

Finding cut-off values. Given $P(X \le k) = p$, find k:

1. Find z such that $\Phi(z) = p$ (using the inverse normal table or technology).
2. Convert back: $k = \mu + z\sigma$.

2.7 Investigations Using the Normal Distribution

This section develops investigative skills with the normal distribution, including:

- estimating population parameters from sample data,
- testing whether a data set is approximately normally distributed (e.g. using a histogram or normal probability plot),
- interpreting results in context.

Checking normality. A data set may be approximately normal if:

- its histogram is roughly bell-shaped and symmetric,
- approximately 68%, 95%, and 99.7% of data lie within 1, 2, and 3 standard deviations of the mean respectively.

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3 Binomial Distributions

3.1 Binomial Probability

A **binomial experiment** has the following properties:

- A fixed number n of trials.
- Each trial has exactly two outcomes: *success* or *failure*.
- The probability of success p is the same on every trial.
- The trials are independent.

If X is the number of successes in n independent trials with probability p of success, then

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}, \qquad r = 0, 1, 2, \ldots, n.$$

Cumulative binomial probability:

$$P(X \leq k) = \sum_{r=0}^{k} \binom{n}{r} p^r (1-p)^{n-r}.$$

3.2 Binomial Distributions

If $X \sim B(n, p)$, we say X follows a **binomial distribution** with parameters n (number of trials) and p (probability of success).

Mean and variance of the binomial distribution:

$$E(X) = np, \qquad \mathrm{Var}(X) = np(1-p).$$

Shape of the distribution:

- When $p = 0.5$, the distribution is symmetric.
- When $p < 0.5$, the distribution is right-skewed (positively skewed).
- When $p > 0.5$, the distribution is left-skewed (negatively skewed).
- As n increases, the distribution becomes more bell-shaped.

Remark 3.1

The binomial distribution is the most important discrete distribution in the HSC syllabus. Make sure you can correctly identify n, p, and r before substituting into the formula.

3.3 Normal Approximations to a Binomial

When n is large, the binomial distribution $B(n, p)$ can be approximated by a normal distribution. Specifically,

$$X \sim B(n, p) \approx N\left(\mu, \sigma^2\right), \qquad \mu = np, \quad \sigma^2 = np(1-p).$$

When is the approximation valid? A common guideline is that both $np \geq 5$ and $n(1-p) \geq 5$.

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Continuity correction. Because the binomial is discrete and the normal is continuous, a continuity correction improves accuracy:

$$P(X = k) \approx P\big(k - \tfrac{1}{2} \le X \le k + \tfrac{1}{2}\big).$$

For example:

$$P(X \le k) \approx P\big(X \le k + \tfrac{1}{2}\big) \quad \text{(normal approximation)}.$$

3.4 Sample Proportions

Given a random sample of size n from a population with true proportion p, the **sample proportion** is

$$\hat{p} = \frac{X}{n},$$

where $X \sim B(n, p)$ is the number of successes in the sample.

Mean and variance of $\hat{p}$:

$$E(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

Approximate distribution. For large n,

$$\hat{p} \approx N\left(p, \frac{p(1-p)}{n}\right).$$

Confidence intervals. An approximate 95% confidence interval for p is

$$\hat{p} \pm 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

4 Advanced Topics

(These topics are outside the HSC Mathematics Extension 2 syllabus but provide valuable enrichment.)

4.1 Monte Carlo Simulations

Monte Carlo simulation is a computational technique that uses random sampling to approximate numerical results. It is particularly useful when a problem is too complex to solve analytically.

Basic idea:

1. Define a random experiment whose outcome is related to the quantity you want to estimate.
2. Simulate the experiment a large number of times (e.g. $N = 10{,}000$ trials).
3. Estimate the quantity as the proportion (or average) of the simulated outcomes.

Estimating π by Monte Carlo. Consider a unit square with an inscribed quarter-circle of radius 1.

- Generate N random points (x_i, y_i) with $x_i, y_i \in [0, 1]$.
- Count the number M of points satisfying $x_i^2 + y_i^2 \le 1$.
- Then $\pi \approx 4M/N$.

Why it works. The probability that a random point in the unit square lies inside the quarter-circle is $\pi/4$, since the area of the quarter-circle is $\pi/4$ and the area of the square is 1.

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4.2 Hypergeometric Distribution

The **hypergeometric distribution** models the number of successes in a sample drawn *without replacement* from a finite population.

Setup. A population of N items contains K successes (and $N-K$ failures). A sample of n items is drawn without replacement. Let X be the number of successes in the sample.

Probability formula:

$$P(X=k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \qquad \max(0,\, n-(N-K)) \le k \le \min(n, K).$$

Mean and variance:

$$E(X) = \frac{nK}{N}, \qquad \mathrm{Var}(X) = \frac{nK(N-K)(N-n)}{N^2(N-1)}.$$

Contrast with binomial. The binomial distribution models sampling *with replacement*; the hypergeometric distribution models sampling *without replacement*. For large N, the two distributions are approximately the same.

Remark 4.1

The hypergeometric distribution arises naturally in quality control, card problems, and ecological sampling.

5 Part 1: Worked Problems

Part 1 contains three sets of problems with detailed solutions. Each solution includes a strategy note, step-by-step working, and a takeaways box.

5.1 Basic

Problem 5.1: Placeholder — Basic Problem 1

Problem 5.1This problem will be added later.

Solution 5.1

Solution to be added.

Takeaways 5.1

- Key insight to be added.

5.2 Medium

Problem 5.2: Placeholder — Medium Problem 1

Problem 5.2This problem will be added later.

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Solution 5.2

Solution to be added.

Takeaways 5.2

- Key insight to be added.

5.3 Advanced

Problem 5.3: Placeholder — Advanced Problem 1

Problem 5.3This problem will be added later.

Solution 5.3

Solution to be added.

Takeaways 5.3

- Key insight to be added.

6 Part 2: Practice and Challenge Bank

Part 2 presents additional problems. Try each problem first, then check the hint if needed.

6.1 Warm-Up Drills

Problem 6.1: Placeholder — Warm-Up Drill 1

Problem 6.1This problem will be added later.

Hint: Hint to be added.

Solution 6.1

Solution to be added.

6.2 Stretch Problems

Problem 6.2: Placeholder — Stretch Problem 1

Problem 6.2This problem will be added later.

Hint: Hint to be added.

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Solution 6.2

Solution to be added.

6.3 Challenge Corner

Problem 6.3: Placeholder — Challenge Problem 1

Problem 6.3This problem will be added later.

Hint: Hint to be added.

Solution 6.3

Solution to be added.

7 Conclusion

Probability distributions are at the heart of statistics and data science. The techniques covered in this booklet — from reading a normal distribution table to applying the binomial model — appear regularly in HSC examinations and in real-world applications such as quality control, risk analysis, and machine learning.

Keep practising with a variety of problems, and remember: understanding the shape and parameters of a distribution is often the key to unlocking the solution.

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